Deadline: March 29, 2026 23:59:59.

Question#1:

In a client-server system for a hypothetical banking application, the clients are ATM machines, and the server is the bank server. Suppose that the client has two operations: withdraw and deposit. Use the broker architectural pattern to document an initial architecture design for this system:

a-Draw a class diagram for this system. Use a client-side proxy to encrypt the data using an encrypt operation, and use a server side proxy to decrypt the data.
b- Discuss how you plan to apply redundancy tactics on the server. Additionally, identify the quality attribute that this tactic will achieve, and discuss any potential side effects of applying it.

Question#2:

You are given a class that processes the purchases for an online store. The class receives calls to retrieve the prices for items from a database, records the sold items, updates the database, and refreshes the webpage.

a-What architectural pattern is suitable for this? Illustrate your answer by drawing a model for the solution, showing the method calls/events.
b-Comment on how applying the pattern will impact the modifiability of the system.
c-Draw a sequence diagram for the update operation.

Question#3:

A "Personal Address Book" application program allows the user to add, delete, search, save and load her contact information. The program separates user (command-line) interface and internal processing subsystem. The internal processing system consists of the following classes: ContactManager (responsible for add and delete operations), ContactFinder (responsible for search operation), and DataManager (responsible for save and load operations).

a-What design pattern can be used to implement the user interface? Explain your answer using class diagram for the entire system.

b- Draw an UML sequence diagram to show the behavioural view of the personal address book program that demonstrates what happens when a user enter a new contact information.

Question#4:

In a system comprising of 3 components: A, B, C. Calling A requires calling B, and calling B requires calling A. Component C is responsible for dissimilar tasks T#1, T#2, and T#3.

a-Comment on the modifiability of this system. What are the problems that you see in this system, and how you solve them?

b-Suppose that T#1 is performed by both component A and C? What does it say about A and C? How you solve this problem?

Question#5:

a-Comment on how you would achieve higher performance for a hypothetical Trent Course Registration System, assuming it utilizes a client-server architecture.

b-Suppose that we want greater availability of the server, discuss what kind of tactics you should use to achieve that.

For all diagrams, use an appropriate modelling tool such as Microsoft Visio. Submit your assignment on blackboard using a pdf file.